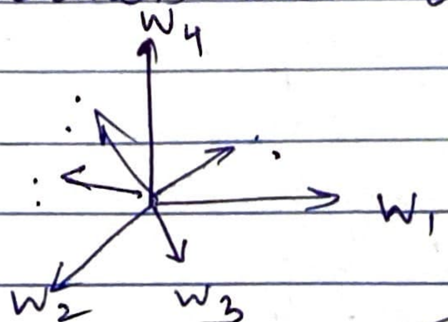
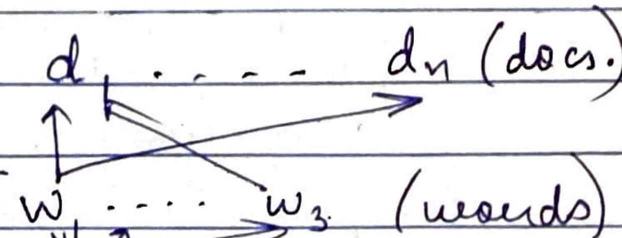
CS6370 | 24^{th} Feb 2021

→ Techniques for Lingual Semantics

(similarity / relatedness)

- how to come up with ways to capture relatedness btw. the words





kind of a bipartite graph

[0, 1] relevance edges that connects words to docs. (tf-idf)

interconnecⁿ tells the similarity

w_1 —— w_3

w'_1 - - - w_3 (similarity matrix)

- Spreading activaⁿ → excite some neurons (nodes) based on the query.
- concept of query expansion ⇐ | w_1 --- w_2 ---

then this layer goes to connect with documents (relevance arcs)

adv: → We might not have explicit info., then also, correct doc. get retrieved. (Semantic Search)

- how to get word relatedness measures
  - but is one single no. enough to capture the entire relatedness of the words? (not enough)
  - are numbers actually good enough for representaⁿ? → the early matrix method is quite naive
  - can we links across docs? superconcept / subconcept: hypernym / hyponym
    part of: meronymy
    synonymy, antonymy
    temporal algebra → reasoning with time
    spatial algebra → reasoning with space.

- MHA → diff. type of nos. are there to capture the info.
- collaborative filtering → kind of recommendaⁿ engine of amazon.
  - there is no content (why?) behind it. It is just 1 person liked, the other person also liked it. It's also based on the user's preference.

- basic methods to estimate the relatedness?

(I Estima" Method)

→ terms (words)

$d_1 \ldots d_n$ → docs

$$\begin{array}{c} t_1 \\ \vdots \\ t_w \end{array} \begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix}$$

t is the dim.
d is the vectors
we might build a space, with flipping d is the dim. & t as its vectors

then find the cosine similarity btw. the words. (Introspective Methods)

same issues
- it will be ltd. to the corpus
→ they are very data hungry,
→ the issue of orthogonality (we are thinking of docs as orthogonal)
→ diff. created by smaller & larger doc sizes.
→ vice-versa leads to circularity problem.
→ docs. are similar if they have the same words. Words are similar if they occur in same docs. Then there is no circularity. But if same is replaced by 'similar', then circularity issue exists. Many algos. like EM, Page rank, etc intend to solve such problems.